

Search titles, descriptions



Welcome to TinyML

Litterature

Hardware

Software

Lesson plan fall 2025

Part 1 | Introduction to TinyML & ML Basics

Part 2 | Embedded Hardware - The Particle Photon2

Part 3 | Data Acquisition

Part 3.1 | Build a data acquisition system

Part 5 | Feature engineering

Part 5.1 | Implement feature calculation on your devic

Part 6 | Introduction to Machine Learning

Part 7 | Introduction to EM Learn, ML on Microcontr

Part 7.1 | Implement a ML model trained on your fei

Part 8 | Introduction to Deep Learning

Part 8.1 | Introduction to Convolutions

Part 9 | Introduction to AiFES, Deep Learning on

Part 10 | Optimisation techniques, Quantisation

Part 11 | Optimisation techniques, Pruning

Part 12 | Anomaly detection

Semester/Exam Project

Hardware

Hardware for TinyML

"Hardware - the parts of a computer you can kick..."

In this course we need hardware to run Machine learning models and interact with sensors.

What hardware is right for you ?
It depends on the application you are to build, and the problem you are trying to solve...

In this course we are using the particle Photon2, it doesn't come with any sensors on board, so you will get an accelerometer and a microphone, to be able to run some of the initial exercises. What sensors are right for you semester project is yet to be found out.



The Photon 2 is compared with a couple of alternative popular hardware devices, often seen in TinyML systems, as they come with a bunch of examples for executing embedded machine learning models - but if you're willing to do a bit of programming on your own, most moderne microcontrollers, e.g. ARM Cortex, some RISC V, and other architechtures will work as well.

In the course we will be working with Audio and Motion, so the Microphone and Inertial Measurement Unit (IMU) or accelerometers are usefull sensors to capture sound and motion.